

AVASH ARYAL

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LinkedIn: www.linkedin.com/in/avasharyal

EDUCATION (100% self-financed)

Bachelor of Science in Electrical Engineering – Micro & Nano Device Concentration

May 2027

Ingram School of Engineering – ABET Accredited Program Texas State

University – San Marcos, Texas

Engineering GPA: 3.97

Honors: Dean's List

RELEVANT COURSEWORK

- Electromagnetics
- Microprocessors
- Microelectronics
- Microelectronics Manufacturing
- Digital Logic
- Circuits I & II
- Signals and Systems
- Solid State Devices
- Hardware Description Languages (HDL)
- VLSI Design

TECHNICAL SKILLS

- Microsoft: Excel, Word, PowerPoint
- Programming/HDL: C++, Python, Verilog (Xilinx Blackboard programming)
- Simulation & Semiconductor Tools: Silvaco ATLAS/DeckBuild TCAD, LTspice, MicroWind
- VLSI: CMOS transistor-level design and layout using MicroWind

EXPERIENCE

Undergraduate Research Assistant (Texas State University)

Spring 2026 – Present

- Performed TCAD simulations in Silvaco ATLAS/DeckBuild to investigate AlScN-based ferroelectric semiconductor devices for memory and in-memory-computing applications.
- Modeled MOSFET transfer/output characteristics and polarization-dependent hysteresis; evaluated doping, device thickness, interface charge, and ferroelectric parameters.
- Analyzed simulated Id-Vg and Id-Vd characteristics using TonyPlot to evaluate AlScN as an active semiconductor channel.

Chemistry Lab Instructor (Chemistry Department Texas State University)

Fall 2024 – Spring 2026

- Troubleshoot laboratory equipment, including LabQuest devices, and developed solutions.
- Instructed and mentored students in laboratory activities.
- Delivered lectures to enhance students' understanding of laboratory concepts.

Junior Technician (Carrollton, TX)

Summer 2024

- Tested and verified laptop functionality from large pallet shipments of second-hand devices.
- Reassembled functional laptops by combining components from broken second-hand devices.
- Ensured quality and performance standards, maximizing usability and reducing electronic waste.

PROJECTS

Senior Design – Renewable Energy Power Conversion with GaN Device

Fall 2026 – Present

- Developing a renewable-energy system integrating solar/wind generation, battery storage, and a DC-DC boost converter controlled through **Arduino-generated PWM**.
- Designing and simulating the converter power stage, gate-drive circuitry, sensing, component sizing, and control behavior using LTspice.
- Project includes fabrication and characterization of a GaN power semiconductor device intended to replace the commercial switching device in the converter.

Embedded C on Xilinx Zynq

GPIO + 7-seg control via memory-mapped I/O.

UART + SPI/I2C; timer-based PWM control and XADC data acquisition.

Semiconductor Manufacturing Lab

Oxidation + oxide thickness metrology on silicon wafers; PVD vacuum diagnostics (pump-down, rate-of-rise).

Photolithography + etching to transfer a Bobcat pattern onto a wafer.

ACTIVITIES and INTERESTS

- **Member, Institute of Electrical and Electronics Engineers (IEEE)**
- **Brain-Engaging Activities**
- **Exploring New Technologies**
- **Exploring Nature and Traveling**

Spring 2024